



Tcl/Tk

Tool Command Language

CSCI330 Programming Languages

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Tool Command Language (Tcl)

Pronounced *tickle*



Overview

- Background and history
- Syntax
- Tk
- Modern Tcl/Tk
- Applications



Background

- Created by John Ousterhout
 - Berkley in 1989
- For use as a basis for integrated command languages.
- Inspired by UNIX shells and Emacs editor.
- Provided as a C library.



Major Tcl Factors

- The language is for commands.
- The language must be programmable.
- The language must permit a simple and efficient interpreter.
- The language must permit a simple interface to C applications.



Example Tcl Program

```
proc fac x {  
    if {$x == 1} {return 1}  
    return [expr {$x * [fac [expr $x-1]]}]  
}
```

```
fac 4 # returns the string "24"
```



Tcl Syntax

Wait, everything is a string?

Always has been.

UNIX/LISP Inspired

- A command consists of one or more fields, space separated.

```
set a {dog cat {horse cow mule} bear}
```

command arguments

- Create variable a with the value
{dog cat {horse cow mule} bear}



Nested Expressions

- ➡ Expressions can be passed as arguments

```
set activity tcl
```

```
set a [format {Elmo's favorite language is %s} $activity]
```



Built in Commands

Control

break, case, continue, eval, for, foreach, if

Variables and Procedures

global, proc, return, set

List Manipulation

concat, index, length, list, range

Expressions

expr

String Manipulation

format, scan, string

File Manipulation

file, glob, print, source

Invoking Subprocesses

exec

Miscellaneous

catch, error, info, time

Figure 1. John K. Ousterhout. 1989. *Tcl: An Embeddable Command Language*. Technical Report. University of California at Berkeley, USA.



Tcl Interpreter

C interface



C Interface

```
typedef int (*Tcl_CmdProc) (ClientData clientData,  
    Tcl_Interp interp, int argc, char *argv[]);1
```

```
Tcl_CreateCommand(Tcl_Interp *interp, char *name,  
    Tcl_CmdProc proc, ClientData clientData);
```

```
int Tcl_Eval(Tcl_Interp *interp, char *cmd);
```

1. The type `Tcl_CmdProc` is created which is a pointer to a function which takes four arguments.

Embedded

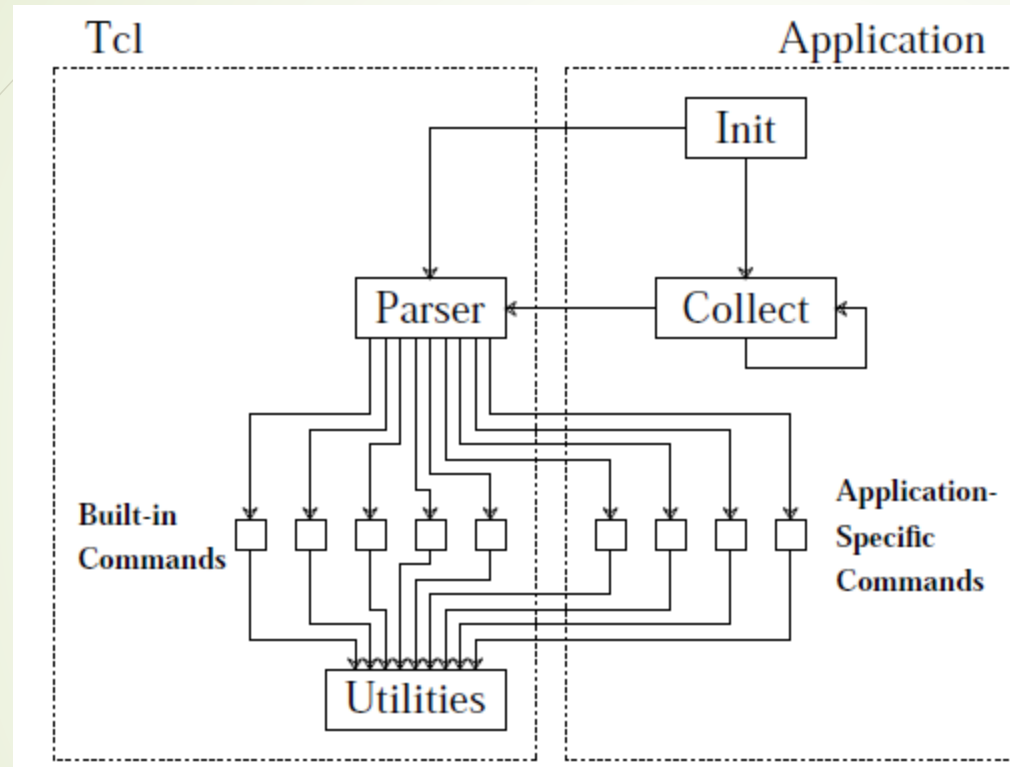


Figure 2. John K. Ousterhout. 1989. *Tcl: An Embeddable Command Language*. Technical Report. University of California at Berkeley, USA.

Control Flow

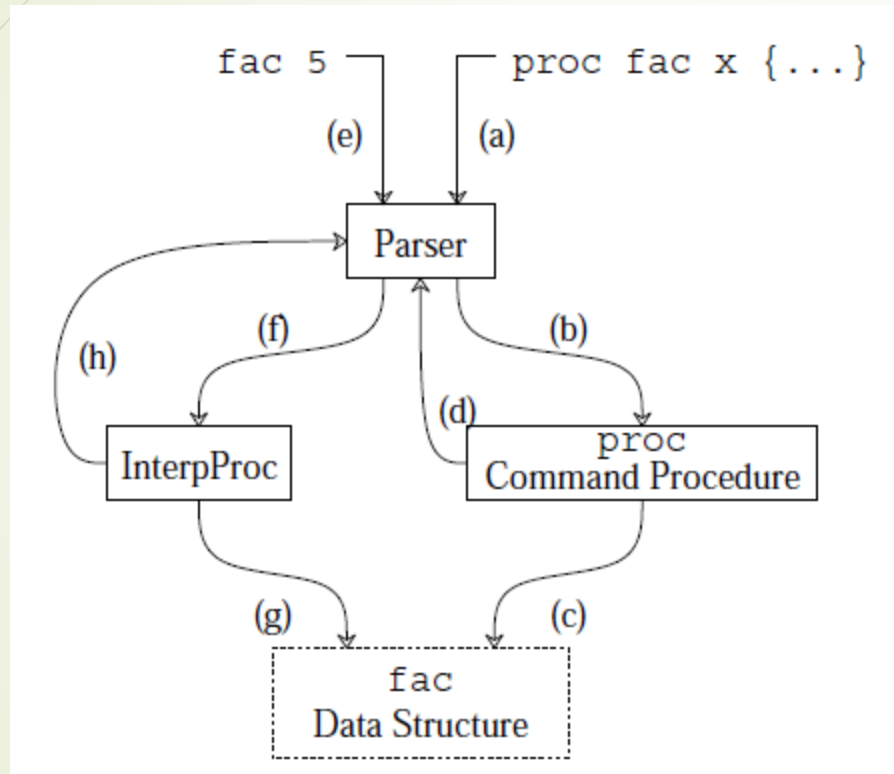


Figure 3. John K. Ousterhout. 1989. *Tcl: An Embeddable Command Language*. Technical Report. University of California at Berkeley, USA.



Tk

Cross platform native GUI



A Natural Next Step

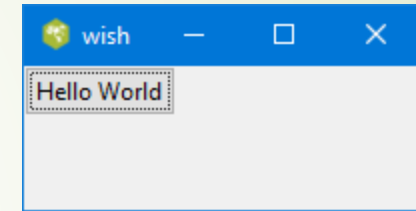
- Tk was developed shortly after Tcl
- Originally for X11 (UNIX) window systems
- Abstracted implementation details for GUI development
 - X11 libraries such as Xt, Xaw and Motif

hello, world²

```
Hello.tcl
```

```
package require Tk
```

```
grid [ttk::button .b -text "Hello World"]
```



2. The first documented `hello, world` program was seen in Kernighan and Ritchie's *The C Programming Language*. An unsettling lack of capitalization and enthusiasm (no exclamation mark!).



Key Contributors

- Scott Stanton, Ray Johnson, Jacob Levy, Brian Lewis, Laurent Demailly
 - Windows and MacOS port (Stanton, Ray)
 - I/O system and sockets (Stanton and Levy)
 - Bytecode compiler, 10x performance (Lewis)
 - Safe-Tcl security model (Levy)
 - Tcl plugins for web support (Levy and Demailly)
 - Jacl and TclBlend for Java (Ousterhout et. Al)



Modern Tcl/Tk

It's everywhere.



Language Adoption

- Python
 - tkinter
- Ruby
 - tk
- Perl
 - Tkx



Python



- tkinter is a standard package included with every modern install of Python.
- Python is installed by default on nearly any modern OS.
- All your computers are Tcl'd.
 - Elmo is pleased.



Tk Evolution – Themed Widgets!

- Dec 2007 saw Tk 8.5.0
- New widgets
 - Trees
 - Combo boxes
 - And more!
- THEMES!
 - Make your application “skinnable”



Open Source

- ➡ Tcl/Tk is now an open source project
- ➡ Tcl/Tk 8.6 (2012)
- ➡ Tcl/Tk 8.6.14 (Feb 2024)
 - ➡ <https://www.tcl.tk/software/tcltk/8.6.html>
- ➡ Tcl/Tk 9.0 (beta)
 - ➡ <https://www.tcl.tk/software/tcltk/9.0.html>



Catching Up: 8.6

- Built in OOP
- Multi threading
- Stackless Evaluation
 - coroutine
 - tailcall
 - yield
 - yieldto
- Database Connectivity with tdbc



Catching Up: 9.0

- 64-bit!
- Zip file mounting!
- Scalable vector graphics!
- Two-finger gesture support!



Educational

Tickling programming curiosity



Introductory Programming Language

- A single data type: String
- Very simple syntax
- Core features included
 - Logic, subprograms, loops
- Exception handling
- Easily create GUI applications



[incr Tcl]

- Package added OOP into Tcl
- Now part of the Tcl standard library
- Easier transition from programming fundamentals to advanced topics



Prototyping

Tickling ideas for rapid development



Redis

- In-memory data store for fast I/O operations with on-disk persistence.
- “...in a 300 line so of code throw-away there was already most of Redis main ideas...from the POV of programming creation process...[it] can already have hints about how to perform future developments...” Salvatore Sanfilippo
- <https://gist.github.com/antirez/6ca04dd191bdb82aad9fb241013e88a8>



VLSI and EDA

- Very Large-Scale Integration (VLSI)
 - Integrated circuits design
- Electronic Design Automation
 - Design and verification of electronic systems



References

- <https://tkdocs.com/tutorial/install.html#helloworld>
- <https://www.tcl.tk/about/history.html>